

Yifeng He

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Curriculum Vitae

Research Interests

My research interests lie in AI-driven software security and engineering, as well as in ensuring the security of AI models. I focus on developing novel AI-based techniques for testing software systems, libraries, and applications.

Education

- 2023-present **Ph.D., Computer Science**, *University of California, Davis*.
- 2019-2023 **B.S., Computer Science, Applied Math**, *University of California, Davis*.

Publications

In Proceedings

- 2026 Ruogu Yang[†], **Yifeng He**[†], Yundi Xu, Yuqing Wei, and Hao Chen. LLM-based invariant testing for software functional bugs. In *IEEE International Symposium on Software Reliability Engineering (ISSRE)*, 2026. [\[link\]](#).
- 2026 **Yifeng He**, Ziyi Tang, and Hao Chen. Content fuzzing for escaping information cocoons on digital social media. In *Findings of the Association for Computational Linguistics (ACL)*, 2026. [\[link\]](#).
- 2026 Hongxiang Zhang, Yuyang Rong, **Yifeng He**, and Hao Chen. Llamafuzz: Large language model enhanced greybox fuzzing. In *ACM/IEEE International Conference on Automation of Software Test (AST)*, 2026. [\[link\]](#).
- 2025 **Yifeng He**, Luning Yang, Christopher Gonzalo, and Hao Chen. Evaluating program semantics reasoning with type inference in system f. In *Neural Information Processing Systems (NeurIPS)*, 2025. [\[link\]](#).
- 2025 **Yifeng He**, Jicheng Wang, Yuyang Rong, and Hao Chen. FuzzAug: Data augmentation by coverage-guided fuzzing for neural test generation. In *Findings of Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. [\[link\]](#).
- 2025 **Yifeng He**, Ethan Wang, Yuyang Rong, Zifei Cheng, and Hao Chen. Security of ai agents. In *International Workshop on Responsible AI Engineering (RAIE)*, 2025. [\[link\]](#).
- 2024 Jiabo Huang, Jianyu Zhao, Yuyang Rong, Yiwen Guo, **Yifeng He**, and Hao Chen. Code representation pre-training with complements from program executions. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024. [\[link\]](#).
- 2024 **Yifeng He**, Jiabo Huang, Yuyang Rong, Yiwen Guo, Ethan Wang, and Hao Chen. UniTSyn: A large-scale dataset capable of enhancing the prowess of large language models for program testing. In *International Symposium on Software Testing and Analysis (ISSTA)*, 2024. [\[link\]](#).
- 2023 Jianyu Zhao, Yuyang Rong, Yiwen Guo, **Yifeng He**, and Hao Chen. Understanding programs by exploiting (fuzzing) test cases. In *Findings of the Association for Computational Linguistics (ACL)*, 2023. [\[link\]](#).
- 2022 **Yifeng He**. Big data and deep learning techniques applied in intelligent recommender systems. In *2022 IEEE 4th International Conference on Civil Aviation Safety and Information Technology (ICCASIT)*, pages 1119–1124, 2022. [\[link\]](#).

Preprints

- 2026 Jicheng Wang, **Yifeng He**, Zili Wang, Hanwen Xing, Arkaprava De, and Hao Chen. Contractbench: Can llm agents preserve observation contracts?, 2026. [\[link\]](#).
- 2026 **Yifeng He**, Yinzhe Zhao, Jicheng Wang, and Hao Chen. Is progressive disclosure all you need for long-context agents?, 2026. [\[link\]](#).
- 2026 **Yifeng He**, Ethan Wang, Jicheng Wang, Xuanxin Ouyang, and Hao Chen. Code generation by differential test time scaling, 2026. [\[link\]](#).
- 2026 Xiangyi Li, Wenbo Chen, Yimin Liu, Shenghan Zheng, Xiaokun Chen, **Yifeng He**, et al. Skillsbench: Benchmarking how well agent skills work across diverse tasks, 2026. [\[link\]](#).

[†]These authors contributed equally (co-first authorship).

- 2024 Hongxiang Zhang, **Yifeng He**, and Hao Chen. Steerdiff: Steering towards safe text-to-image diffusion models, 2024. [\[link\]](#).
- 2024 Jicheng Wang, **Yifeng He**, and Hao Chen. Repogenreflex: Enhancing repository-level code completion with verbal reinforcement and retrieval-augmented generation, 2024. [\[link\]](#).

Services

Organizing Committee

- 2026 The First Workshop on Agent Skills (Agent Skills '26)

Reviewer / Program Committee

- 2026 ACM Conference on AI and Agentic Systems (CAIS), Main & Demo
- 2026 Conference on Language Modeling (COLM)
- 2025,2026 International Conference on Learning Representations (ICLR)
- 2026 Neural Information Processing Systems (NeurIPS)
- 2026 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)
- 2025,2026 International Conference on Multimedia (MM)
- 2025 International Conference on Software Engineering Advances (ICSEA)
- 2025 Association for Computational Linguistics LLM Security Workshop (ACL-LLMSEC)

Artifact Evaluation Committee

- 2026 IEEE Symposium on Security and Privacy (S&P)
- 2025 Conference of the ACM Special Interest Group on Data Communication (SIGCOMM)

Awards

- 2023 **Citation for Outstanding Performance**, Department of Mathematics, UC Davis
- 2019-2022 **Dean's Honor List**, College of Letters and Science, UC Davis

Industry Experiences

- 06/2026- **Applied Scientist Intern**, *Amazon Web Services (AWS)*, Kiro Science.
- 09/2026
 - Research on making agents smarter on analyzing code
- 07/2021- **Software Development Engineer Intern**, *ByteDance (Xigua Video)*, Creator Experience Team.
- 08/2021
 - Created a data cleaner with ORM to maintain the size and readability of the online data settlement table
 - Created the offline flow of Medium-Length Video Encouragement Project for weekly data calculation
 - Built the interface for the frontend of the web and mobile app to display the data visualization
- 04/2021- **Software Development Engineer Intern**, *ByteDance (Toutiao)*, Creator Income Platform.
- 07/2021
 - Used microservice tech to connect parts of the author income settlement business
 - Transformed author-relation data architecture design from relational database (SQL) to graph database (Gremlin) to allow better efficiency for the business model
 - Refactored the income calculation control process within Python 3